

LevelUp Study Research

Why LevelUp Study uses tests to help students learn (The testing effect)

Most students revise by rereading their notes and lecture slides over and over. This feels productive because the material becomes familiar, but research proves it is actually one of the least effective ways to prepare for exams.

Roediger and Karpicke [1] conducted a study where one group of students repeatedly studied a passage, whilst another group studied it once and then took multiple tests. After one week, the students who took tests remembered 61% of the material, and the students who only studied remembered just 40%. This was despite the study group reading the passage over fourteen times, more than four times as many readings as the test group. This is called the 'testing effect' and its clearly demonstrated in the study as the act of retrieving information from memory is what makes learning stick, not rereading.

LevelUp Study also has the ability to provide a short summary as a refresher, enough to bring key points back to mind, but the real learning happens when students are forced to retrieve that information. Whether it is multiple choice, fill-in-the-blank, or written answer questions, the act of pulling information out of their own memory is what strengthens retention and prevents forgetting.

Why LevelUp Study uses gamification to help students learn

Students do not fail exams because they are incapable, they often fail because learning and revision can be unengaging and easily put down till the last day. Gamification research directly addresses this problem. Majuri, Koivisto and Hamari [2] conducted a comprehensive review of 128 empirical studies on gamification in education and found that 71% of studies reported positive results.

The review also identified which gamification elements are most effective. Points, scores, levels, progress bars, and leaderboards categorised as "achievement and progression", were the most commonly implemented and the most positively reported [2]. LevelUp Study uses points, scores and a dynamic progress bar. These elements are referred to as progression-based gamification and are used to increase both motivation and performance.

Importantly, the research shows that gamification produces positive effects on the outcomes that matter most for LevelUp Study. Students report higher enjoyment and engagement as well as improved grades, faster task completion, and better learning progression [2]. LevelUp Study uses gamification for one simple reason, it makes hard work feel more engaging, and the evidence proves that when students are engaged, they learn and perform better.

Why LevelUp Study uses multiple-choice tests to help students learn

Some people assume multiple-choice tests are only good for assessment, that they measure what you know but do not actually help you learn. Research proves this assumption is wrong.

Butler [3] reviewed decades of studies on multiple-choice testing and concluded that "retrieving information in response to a test question strengthens memory, leading to better retention of that information over time." In other words, the act of answering a multiple-choice question is itself a learning event. Students are not just picking an answer, they are actively pulling information out of their memory, and that process makes the memory stronger. This matters for LevelUp Study because the goal is not to make students better at multiple-choice questions, the goal is to help them learn and perform better on their actual exams.

Why LevelUp Study uses fill-in-the-blank questions to help students learn

Fill-in-the-blank tests require students to retrieve specific words or phrases from memory. This process is known as cued recall, and research shows that retrieving information strengthens long-term retention.

Hinze and Wiley [4] conducted three experiments in which students read science texts and then either completed fill-in-the-blank questions or reread the material. When tested again after two days and after seven days, students who had completed the fill-in-the-blank tests performed significantly better on the exact information they had previously retrieved [4].

These findings support LevelUp Study's inclusion of fill-in-the-blank questions. Multiple-choice questions are effective, but the correct answer is visible on screen, allowing students to recognise the answer without retrieving it from memory. Fill-in-the-blank removes this support, requiring students to actively recall the term, name, or number themselves. This act of retrieval drives the testing effect, where successfully recalled information becomes more likely to be remembered later [4].

Why LevelUp Study uses written answer questions to help students learn

Multiple choice and fill-in-the-blank questions are effective, but the most powerful form of retrieval practice is free recall. Butler and Roediger [5] tested this directly in a simulated classroom. Students watched lectures and then either took a short written answer test, took a multiple choice test, or did nothing.

One month later, the students who had written short answers remembered significantly more than every other group. They scored 47% on the final test, compared to 36% for multiple choice, and just 20% for students who did nothing [5].

The reason writing works is that it forces students to organise their knowledge, construct a response, and identify gaps in their own understanding. Butler and Roediger [5] explain this as "greater depth or difficulty in retrieval" and confirm that this difficulty is precisely why writing produces stronger long-term retention. The evidence is clear, if you want students to remember something for the long term, make them write about it.

References

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